

Postdoctoral Associate, Institute of Behavioral Science, University of Colorado, Boulder
Ph.D. Geography, Arizona State University
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Education

Doctor of Philosophy, Geography
Arizona State University · Tempe, AZ · 2024

- Dissertation: The geography of inequality in STEM: migration, gender, and economic disparities.
- Advisors: Dylan Connor (co-chair, Stanford/ASU); Wei Li (co-chair, ASU); Mark Ellis (UW)
- GPA: 4.0/4.0

Master of Science, Urbanisation and Development
London School of Economics and Political Science · London, UK · 2017

- GPA: 3.7/4.0

Bachelor of Arts, Geography
University of Washington · Seattle, WA · 2015

- GPA: 3.6/4.0

Research Interests

Urban Spatial Demography and Hyper-Local Population Health:
Build longitudinal, high-resolution health and population data by linking health records, socioeconomic, built environment, and demographic microdata. Develop scalable spatial-epidemiologic and GeoAI frameworks to quantify how economic dynamics, built environments, and community resources shape health inequities over the life course.

Decision-ready GeoHealth for Science and Policy:
Bridge GIScience and population health to produce reproducible, policy-facing GeoHealth analytics. Partner with public agencies, healthcare systems, and NGOs to monitor health trends, identify high-risk neighborhoods, and evaluate place-based interventions using privacy-aware, longitudinal geospatial workflows.

Skills

Data Analysis: R, SAS, Python, SQL, Tableau, SPSS, FSQCA, MS excel, Machine Learning, Bayesian Causal Inference
Spatial Analysis: ArcGIS Series, Google Map API, Geoda, FRAGSTAT, Spatial Econometric Models, Spatial Temporal Bayesian Models, Community Detection Models
Language: Mandarin Chinese (proficient), English (proficient), Greek (basic), Spanish (basic)

Research Experience

University of Colorado Boulder· CU Population Center & Institute of Behavioral Science
Postdoctoral Associate · Boulder, CO · November 2024 to Present

- **CU Population Center Seed Grant (Role: PI). Neighborhood Mortality Regimes Across the Life Course: Community Wealth, Family Stability, and Youth Mortality**
 - Directed a spatial demography pilot study investigating highly localized youth mortality regimes, integrating geocoded death certificates with novel tract-level community wealth databases.
 - Developed multilevel spatial models, Bayesian Causal Inference (in R/Python) to analyze the how community wealth and family stability in determine neighborhood disadvantages and pediatric health risks.
 - Established an expansion-ready, small-area mortality data infrastructure, providing proof-of-concept for localized socio-environmental interventions and future multi-state R01/ESRC grant applications.
- **National Institute on Aging. R01. Addressing the Midlife Mortality Crisis: Place-Based Modeling, Trend Analysis and Policy Interventions**
 - Developed large-scale geospatial health datasets with over 10 million confidential mortality records from 16 state governments using R, GeoDa, Python and SAS.

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- Created machine-learning longitudinal models to monitor mid-life mortality trends and demographic and socioeconomic changes across American neighborhoods, understanding place-based health determinants for the mid-aged population.
- Analyzed regional and local determinants of mid-life mortality in the US by utilizing Bayesian multinomial models, multi-level models and spatial autocorrelation models.
- Cleaned, extracted and analyzed 2 million business establishment records from the National Establishment Time Series (NETS) Datasets.
- Constructed a working database integrating business records, census data and health records to analyze the health impacts of business dynamics using Geographically Weighted Regression.
- Lead conference presentation and drafting manuscripts.

Acumen LLC / SPHERE Institute

Data & Policy Analyst • Burlingame, CA • November 2022 to May 2024

- Clean, process, and analyze large health datasets, including over 20 million Medicaid, and nursing facility records.
- Evaluate nursing home service quality and medical insurance program integrity.
- Developed sequence analysis method with clinicians and policy experts to detect fraud claims and drug abuse.
- Utilized SAS EBI, streamlined analysis of longitudinal data from the Integrated Data Repository, managed, and scrutinized insurance claims following beneficiary death / posthumous claims.
- Drafted policy reports and visualizations for regular client meetings.
- Identified regional patterns in prevalent health conditions and high-risk populations by analyzing confidential Medicaid and Medicare claims data using SAS, R, SQL, multiple regression, and frequency charts.
- Presented comprehensive reports to clients, including the Centers for Medicare & Medicaid Services (CMS), Center for Program Integrity, and Nursing Home Quality Initiative.

Arizona State University • Spatial Analysis Research Center

Graduate Research Associate (Part-time) • Tempe, AZ • August 2018 to November 2024

- **National Institute of Health. R21. Trends, transitions, and well-being in small-town America.**
 - Established typologies of longitudinal demographic and socioeconomic changes in small towns of the US using sequence analysis, machine-learning clustering, and geospatial analysis in R, Python, and ArcGIS.
 - Mapped and modeled community change and health risk using geospatial clustering, multinomial logit models and spatial autocorrelation models in R and ArcGIS.
 - Presented research findings at the National Association of Regional Science conference.
 - Drafted annotated bibliography for studies of longitudinal community change.
- **National Science Foundation: Humans, Disasters, and the Built Environment program. The creeping disaster along the coast: Built environment, coastal communities and population vulnerability to sea level rise.**
 - Analyzed built-environment features and developed urban landscape metrics of 709 commuting zones in the US with remote sensing images and Ztrax dataset using python and ArcGIS.
 - Predicted regional socioeconomic change in relation to build-environment changes, using machine learning feature selection, regularization and causal inference techniques in Python and R.
- **COVID-19 and a World of Ad Hoc Geographies book project.**
 - Led a three-person team, developed the research design, and oversaw data analysis.
 - Developed a community-level race-specific social vulnerability measure to predict racial inequality and identify communities at risk of COVID-19 using ACS PUMS data.
 - Created California state-wide hot-spot maps for anti-Asian racism incidents using Google Map Geolocation API and online survey from STOP AAPI HATE NGO.
 - Drafted the introduction, method, findings and discussion sections of 1 peer-reviewed book chapter.
 - Delivered 4 in-person and online presentations to STOP AAPI HATE NGO, Arizona State University and Eastern Mennonite University, and Association of American Geographer national conference.

Other Professional Experience

Cushman & Wakefield

Assistant Manager • Guangzhou, China • December 2017 to July 2018

- Led a six-person team in delivering strategic consulting services to city governments and private corporations.

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- Advised on eight urban development projects, including Yuexiu retirement community development, Haikou Free Trade Area TOD urban complex, and Six Flags Park China.

Peer-reviewed Publications

- Xie, S.**, & Connor, D. (2026). Stemming the tide of inequality? Immigration, gender, and spatial income disparities in today's leading technological sectors. *Annals of the Association of American Geographers*. DOI:10.1080/24694452.2026.2648332
- Connor, D., **Xie, S.**, et al. (2025). Sprawling cities fuel inequality across generations. *PNAS Nexus*, 00, pgae587.
- Connor, D., **Xie, S.**, et al. (2024). Poverty dynamics and social mobility in rural America. *Population Space and Place*, e-2802.
- Xie, S.** (2023). Global education trajectories and inequality: STEM workers From China to the US. *Journal of Ethnic and Migration Studies*, 49(18), 4699-4721.
- Xie, S.**, Li, W., & Tan, Y. (2021). Vulnerability, Infection, and Anti-Discrimination among Asian Americans and COVID-19: Explorations into Race and Health Intersections in California. In S.D. Brunn (Ed.), *COVID-19 and a World of Ad Hoc Geographies*. New York: Springer.

Manuscripts Under review

- Xie, S.**, Gutmann, M., et al. (2026). Mapping Neighborhood Patterns in the U.S. Mid-Life Mortality Crisis: a Sequence Analysis. *Social Science and Medicine Population Health*. (Under Review). Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6680378
- Xie, S.**, Parker, E., et al. (2026). Policy Liberalism vs Neoliberalism: How State Policy Buffers Deindustrialization and the Rural Mortality Penalty. *Rural Sociology*. (extended abstract accepted)
- Xie, S.**, Leyk, S., et al. (2026). The Multi-Scalar Drivers of Neighborhood Mid-Life Mortality Trajectories in the United States. *Proceedings of the National Academy of Science*. (Under Review)
- Connor, D., **Xie, S.**, et al. (2026). Community Wealth Shapes Cognitive Health Among Older Adults. *Science Advances*. (Revise & Resubmit). Available at ResearchGate, DOI:10.31235/osf.io/fpcr5_v1
- Kemeny, T., **Xie, S.**, et al. (2026). Spatial Wealth Inequality in the United States: Theory and Evidence. *Journal of Economic Geography*. (Under Review). Available at SocArXiv: <https://osf.io/preprints/socarxiv/r4sg9>
- Gutmann, M., **Xie, S.**, et al. (2025). Big Data for Health Research: Understanding Working-Age Mortality at Very Fine Spatial Scales. *Scientific Data*. (Under Review).
- Xie, S.**, & Connor, D. (2025). Highly skilled immigrants make outsized economic contributions in lagging regions. *Humanities and Social Science Communications*. (Under Review)

Other Publications

- Xie, S.** (2025). Crosswalks between Commuting Zones and Historical Geographies (1990–2020) [Data set]. University of Colorado Boulder. <https://doi.org/10.25810/5XP7-YE36>
- Xie, S.** (2025). 100*100m Population Grid (2020 Census) [Data set]. University of Colorado Boulder. <https://doi.org/10.25810/PB2D-P578>
- Xie, S.** (2025). GEO-NSCG 2021 5-Year Estimates [Data set]. University of Colorado Boulder. <https://doi.org/10.25810/5D5P-FX20>
- Xie, S;** Li, W (2019), New representation or new misrepresentation? Critical reflection of Crazy Rich Asians. *Model Minority Magazine at Dartmouth College*, inaugural issue,

Presentations

- 2026, **Xie, S;** What drives Neighborhood Patterns in the U.S. Mid-Life Mortality Crisis: A Multinomial Spatial Analysis, *Annual conference of the American Association of Geographers*
- 2025, 2026. **Xie, S;** Mapping Neighborhood Patterns in the U.S. Mid-Life Mortality Crisis: A Sequence Analysis, *Annual North American Meetings of the Regional Science Association, Webinar Series of the Sequence Analysis Association*
- 2025, **Xie, S;** Immigration and America's new wealth frontier, *Annual conference of the American Association of Geographers, Stanford Center for Advanced Study in the Behavioral Sciences Geowealth workshop*
- 2022, **Xie, S;** Pathway to Success: education trajectories and inequality in STEM, *Annual conference of the American Association of Geographers*
- 2022, **Xie, S;** The geography of foreign education: imputing location of foreign-born STEM workers in the US, ASU Spatial Analysis Research Center (SPARC) presentation series

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2022, **Xie, S**; Pathway to Success: education trajectories and inequality in STEM, *Annual conference of the American Association of Geographers*
2022, 2021, **Xie, S**; Li, W; Covid-19 and Anti-Asian racism, invited talks at *ASU School of Social Transformation, Eastern Mennonite University Center for Justice and Peacebuilding*
2021, **Xie, S**; Occupational selectivity and migration-gender wage inequality in STEM. *Race, Ethnicity, and Place Conference*, the Middle Atlantic Division of the American Association of Geographers (MAD-AAG) joint meeting.
2021, **Xie, S**; Different major: geography. Podcast at ASU International Student Service Center
2019, **Xie, S**; Lacey, E. How can sharing personal narratives of migration help us find connectedness in our differences? Invited talk at ASU Graduate College 80th Anniversary
2019, **Xie, S**; Cooke, A; Connor, D. Employment Dynamics and Long-term Neighborhood Change: Buffalo, NY 1940 to 2017. 66th Annual North American Meetings of the *Regional Science Association International*

Works in Progress

Xie, S (with Gutmann, M., Leyk, S. and Berg, A) Mid-life mortality crisis and local determinants of health
Xie, S (with Gutmann, M., Leyk, S. and Sun, Y.) Business dynamics and the death of despair
Xie, S (with Connor, D, Storper, M. Kemeny, T. Jang, J.) Racial wealth inequality in the US
Xie, S (with Connor, D, Storper, M. Kemeny, T.) The US housing crisis and its impact on long-term wealth inequality
Xie, S (with Connor, D, Suss, J. Kemeny, T.) Geography of immigrant wealth in the US
Xie, S (with Connor, D.) Where are the “World’s Best and Brightest”: foreign-born workers, and STEM labor markets of the US
Xie, S (with Cooke, A; and Connor, D;) Employment Dynamics and Long-term Neighborhood Change: Buffalo, NY 1940 to 2017

Consulting Reports

Xie, S. Panishkavili.N (2023). “Health condition trend among Medicaid beneficiaries in the states of New York and Pennsylvania”. Consultancy report to Center of Medicaid and Medicare Services. Acumen LLC/SPHERE Institute.
Xie, S (2023). “Antipsychotic drug abuse monitoring using time series analysis”. Consultancy report to Nursing Home Quality Initiative. Acumen LLC/SPHERE Institute.
Xie, S. Byrne, E. (2023). “Evaluating State program integrity and database for detecting posthumous insurance claims”. Consultancy report to Center of Medicaid and Medicare Services. Acumen LLC/SPHERE Institute.
Xie, S (2018), “Urban redevelopment and rural preservation in Guangdong Province”, Litanba renewal project, consultancy report to Municipal government of Meizhou and Trust King urban renewal group, Cushman & Wakefield, Strategic Consulting, Guangzhou, China
Xie, S. Chen, V, Zheng, H. et. Al. (2018), “Recreational and healthcare themed town development” in urban development and real estate investment planning, consultancy report to Riverside Group and Six Flags Entertainment Corporation, Cushman & Wakefield, Strategic Consulting, Guangzhou, Beijing, Chengdu, Hangzhou, Shanghai, China
Xie, S. Chen, V. (2017), “Continue care and recreational community in Southern China”, consultancy report to Yuexiu Holding, Ltd. Cushman & Wakefield, Strategic Consulting, Guangzhou, China

Honors and Grants

2026	CUPC Seed Grant (Supported by NIH/NICHD)	Neighborhood Mortality Regimes Across the Life Course: Community Wealth, Family Stability, and Youth Mortality (PI , \$ 12,909)
2026	Institute of Behavioral Science Summer RA Award	Modeling Landscape Transitions and Emerging Fire Risk in the United States – A Colorado case study (Faculty Advisor , \$2,000)
Pending	Russell Sage Foundation	The changing geography of racial wealth inequality, 1960-2022 (Co-I)
Pending	NIH/NICHD	Youth Mortality, Mobility and Community Wealth (PI , to be submitted in October 26)
2022	ASU GPSA	Travel grant (\$350)
2021	ASU Graduate School	Travel award (\$1000)
2019	ASU SGSUP	Research symposium presentation award (2nd prize)

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2015	University of Washington	Study abroad travel grant (\$5000)
2012–15	University of Washington	Dean’s List

Teaching Experience

University of Colorado Boulder (Institute of Behavioral Science)

- Doctoral Student Mentor — Faculty Mentor • Boulder, CO • March 2026 to Present
- Mentored a Geography PhD researcher through the ideation and development of a successful competitive funding proposal, resulting in the award of an IBS Summer Research Grant.
 - Designed and delivered one-on-one methodological training in advanced machine learning algorithms (sequence analysis) to build the student’s independent quantitative research capacity.
 - Established a structured doctoral supervision framework—including project timelines, weekly progress meetings, and milestone tracking—to ensure rigorous research execution.
 - Guided the translation of research findings into academic and public-facing outputs, leading the student toward a targeted Fall academic presentation and the development of research exhibition materials.

Arizona State University

- Graduate Teaching Associate • Tempe, AZ • August 2018 to November 2022
- Facilitated the following courses: Urban Economic Development (2018, Graduate seminar), World regions (2018-2022), Population Geography (2019-2022), International Crisis (2020-2022).
 - Designed, led, and assessed guest lectures, class activities, assignments, and projects for population growth theories, population geography, and regional geography of East Asia.
 - Designed instruction plan, class presentation, instruction video and public lecture for population data access and analysis.
 - Mentored two undergraduate honor’s theses.

Seattle Central College

- Tutor • Seattle, WA • January 2015 to December 2015
- Offered career consultation services, providing guidance on job search, resume building, and interview preparation.
 - Assisted 13 students with their English reading, writing, and speaking through personalized tutoring sessions.

Service Experience

Ongoing	Journal Reviewer	BMC Public Health, Demography, Sociology of Race and Ethnicity, Humanities and Social Sciences Communications, Computational Economics
Ongoing	Postdoc panelist	CU Boulder IBS Professionalization Seminar
2026	Session Chair	Geography of Mortality I & II sessions at the American Association of Geographers 2026 Annual Meeting.
2025	Session Discussant	<i>Annual North American Meetings of the Regional Science Association</i>
2020–2021	COVID and anti-racism focus group	ASU SGSUP JEDI committee
2019	Conference Assistant	American Association of Geographers
2018–2019	International student coordinator	ASU SGSUP Grad student committee

Affiliations

Population Program, Institute of Behavioral Science, Population Association of America, Association of American Geographers, Regional Science Association International, Sequence Analysis Association